Project Proposal Template

Project: Global shipping disruption Analytics

Objective:. Analyze the impact of crises on shipping traffic and the estimated cargo volume at the main maritime chokepoints worldwide.

Description: This project analyzes how crises affect ship traffic and cargo volume at the main maritime chokepoints, and how long it takes for shipping to return to normal. It covers global shipping disruptions from 2019 to the present, using IMF PortWatch data.

Tools & Design: Data warehouse with Bronze, Silver, and Gold layers and a star schema (fact and dimension tables) in SQL Server, EDA in Python, and a Power BI dashboard.

Technologies to Use: SQL Server, Python, Power BI.

Team Members & Responsibilities:

1- **Shahd Waheed Eddin Saad:** SQL Server: schema design, relationships, and part of Gold. Python: data quality checks. Power BI: data modeling and filter propagation testing.
2- **Basmala Mahmoud Elsayed:** SQL Server: loading raw data into Bronze and validation. Python: feature engineering. Power BI: DAX measures, KPIs, and checking the calculations.
3- **Tasbeeh Mohamed Hassan:** SQL Server: cleaning in Silver (missing values and duplicates). Python: univariate EDA (distributions and outliers). Power BI: Executive Overview page.
4- **Aisha Taher Hussien:** SQL Server: fact and dimension tables in Gold. Python: bivariate and multivariate EDA (relationships). Power BI: analytical page with drill-down and filters.
5- **Abrar Mohamed Abdelghani: SQL** Server: validation queries, data quality review, and business queries. Python: EDA on an independent part of the data and business insights. Power BI: analytical page and testing the interactions and consistency of results.

Week 1: *SQL Server & Data Warehouse (Design and Bronze)*

- **Shahd Waheed Eddin Saad:** Design the schema and define the relationships.
- **Basmala Mahmoud Elsayed:** Load the raw data into Bronze and validate it.
- **Tasbeeh Mohamed Hassan:** Start the cleaning in Silver (missing values and duplicates).
- **Aisha Taher Hussien:** Design the fact and dimension tables for Gold.
- **Abrar Mohamed Abdelghani:** Prepare the SQL validation queries.

Deliverables:

- Schema design with relationships.
- Raw data loaded into Bronze and validated.
- Design of the fact and dimension tables.
- SQL validation queries.

Week 2: *SQL Server & Data Warehouse (Silver and Gold)*

- **Shahd Waheed Eddin Saad:** Implement part of Gold.
- **Basmala Mahmoud Elsayed:** Validate the data after the Silver and Gold layers are built.
- **Tasbeeh Mohamed Hassan:** Finish the cleaning in Silver (missing values and duplicates).
- **Aisha Taher Hussien:** Implement the fact and dimension tables in Gold.
- **Abrar Mohamed Abdelghani:** Run the SQL validation queries, review the data quality, and write the business queries.

Deliverables:

- Cleaned data in Silver (missing values and duplicates handled).
- Gold layer with fact and dimension tables.
- Data quality review.
- Business queries.

Week 3: *Python*

- **Shahd Waheed Eddin Saad:** Data quality checks and fixing the remaining issues.
- **Basmala Mahmoud Elsayed:** Feature engineering (creating derived columns).
- **Tasbeeh Mohamed Hassan:** Univariate EDA (distributions and outliers).
- **Aisha Taher Hussien:** Bivariate and multivariate EDA (analyzing relationships).
- **Abrar Mohamed Abdelghani:** EDA on an independent part of the data and extracting business insights.

Deliverables:

- Data quality checks and fixes.
- Derived columns (feature engineering).
- Univariate EDA (distributions and outliers).
- Bivariate and multivariate EDA (relationships).
- Business insights from the EDA.

Week 4: *Power BI*

- **Shahd Waheed Eddin Saad:** Data modeling, relationships, and testing filter propagation.
- **Basmala Mahmoud Elsayed:** DAX measures, KPIs, and checking the calculations.
- **Tasbeeh Mohamed Hassan:** Design a complete Executive Overview page.
- **Aisha Taher Hussien:** Design a complete analytical page with drill-down and filters.
- **Abrar Mohamed Abdelghani:** Design a complete analytical page, then test the interactions and the consistency of the results.

Deliverables:

- Data model with tested relationships.
- DAX measures and KPIs with verified calculations.
- Executive Overview page.
- Two analytical pages with drill-down, filters, and tested interactions.